

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Case Study Analysis

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Q15. Case Study 15

Develop an appropriate solution for the given Generative AI/LLM scenario. Your answer should include analysis, justification, suitable model selection (GPT/BERT/RLHF/LoRA/PEFT where applicable), implementation, approach, expected outcome, and limitations. Topic: Medical Q&A assistant

Solution:

Case Study: Generative AI-Based Medical Q&A Assistant

Case Study

Background

A large multi-specialty hospital receives thousands of patient inquiries every day through its website, mobile application, and customer support center. Patients frequently ask questions about symptoms, medications, diseases, appointment scheduling, vaccination, laboratory tests, and post-treatment care. Responding to these repetitive queries manually increases the workload of healthcare professionals and leads to longer response times.

To improve healthcare accessibility and reduce the burden on medical staff, the hospital plans to develop a **Generative AI-powered Medical Q&A Assistant**. The assistant will provide accurate, evidence-based answers to general medical questions, guide patients to appropriate healthcare services, and recommend consulting a doctor for emergencies or complex medical conditions. The system will not replace physicians but will act as an intelligent virtual assistant available 24/7.

1. Analysis

The proposed Medical Q&A Assistant should satisfy the following requirements:

- Understand natural language questions from patients.
- Provide medically accurate and easy-to-understand responses.
- Support multiple medical topics such as symptoms, diseases, medicines, vaccination, nutrition, and preventive care.
- Maintain patient privacy and data security.
- Escalate emergency cases to healthcare professionals.
- Avoid generating misleading or unsafe medical advice.

The system should also integrate with the hospital's electronic health record (EHR) system and medical knowledge databases to provide reliable responses.

2. Justification

Traditional rule-based chatbots rely on predefined questions and answers, making them unsuitable for handling diverse medical queries.

A **Generative AI model based on GPT** is more appropriate because it can:

- Understand complex patient questions.
- Generate natural and conversational responses.
- Summarize medical information.
- Answer follow-up questions.

- Support multilingual communication.

Since medical information is sensitive, the AI should retrieve information from trusted clinical guidelines rather than relying solely on its pre-trained knowledge.

3. Suitable Model Selection

Primary Model: GPT

GPT is selected because it is designed for natural language generation and conversational AI.

Reasons:

- Excellent conversational ability.
- Generates detailed explanations.
- Handles follow-up questions naturally.
- Suitable for virtual assistants.

Supporting Model: BERT

BERT can be used for:

- Medical intent classification.
- Disease category identification.
- Medical entity recognition.
- Patient query classification.

Example:

Patient asks:

"I have chest pain and difficulty breathing."

BERT classifies the query as **Emergency** before sending it to GPT.

RLHF (Reinforcement Learning from Human Feedback)

RLHF is applied to improve response quality by incorporating feedback from healthcare professionals.

Benefits:

- Reduces hallucinations.
- Improves medical accuracy.

- Encourages safe recommendations.
- Produces more patient-friendly responses.

LoRA (Low-Rank Adaptation)

LoRA enables efficient fine-tuning of GPT using hospital-specific medical datasets.

Advantages:

- Faster training.
- Lower GPU memory usage.
- Lower computational cost.
- Easy updates when medical guidelines change.

PEFT (Parameter-Efficient Fine-Tuning)

PEFT further reduces computational requirements by updating only a small portion of model parameters.

Benefits:

- Faster deployment.
- Cost-effective.
- Suitable for hospitals with limited computing resources.

4. Implementation Approach

Step 1: Data Collection

Collect trusted healthcare information from:

- WHO Guidelines
- CDC Publications
- Hospital Treatment Protocols
- Medical Textbooks
- Drug Databases
- Frequently Asked Patient Questions

Step 2: Data Preprocessing

- Remove duplicate records.
- Standardize medical terminology.
- Remove sensitive patient information.
- Convert documents into searchable knowledge chunks.

Step 3: Model Training

- Start with a pre-trained GPT model.
- Fine-tune using LoRA or PEFT on hospital-specific medical documents.

Apply RLHF using feedback from doctors.

Step 4: Query Processing

Patient Question

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BERT identifies:

- Intent
- Medical category
- Urgency level

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GPT generates the response

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Knowledge retrieval verifies medical facts

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Safety filter checks for harmful advice

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Final response sent to patient

Step 5: Deployment

Deploy the assistant through:

- Hospital Website
- Mobile Application
- WhatsApp Chatbot
- Voice Assistant

- Customer Support Portal

5. Expected Outcome

After deployment, the hospital expects:

- 24/7 patient support.
- Faster responses to medical queries.
- Reduced workload for doctors and nurses.
- Improved patient satisfaction.
- Consistent evidence-based answers.
- Better healthcare accessibility.
- Automatic escalation of emergency cases.
- Reduced waiting time in call centers.

6. Limitations

Despite its advantages, the Medical Q&A Assistant has several limitations:

- It cannot replace professional medical diagnosis.
- Incorrect or outdated medical knowledge may affect responses if not regularly updated.
- GPT may occasionally generate inaccurate information (hallucinations).
- Patient privacy and data security must be strictly maintained.
- Complex medical cases still require consultation with licensed healthcare professionals.
- Continuous monitoring and expert validation are necessary to ensure safety and reliability.

Overall Solution

The proposed **Generative AI Medical Q&A Assistant** combines **GPT** for natural language response generation, **BERT** for medical query understanding and classification, **RLHF** to improve safety and response quality through expert feedback, and **LoRA/PEFT** for efficient fine-tuning using

hospital-specific medical data. By integrating these technologies with trusted medical knowledge sources and safety checks, the hospital can provide accurate, accessible, and scalable patient support while ensuring that critical or emergency cases are directed to healthcare professionals. This solution improves operational efficiency, enhances patient experience, and supports better healthcare delivery, while recognizing that AI serves as an assistive tool rather than a replacement for qualified medical practitioners.